

Construction of a Technical Tariff Plan for a Belgian Motor Third-Party Liability Portfolio

Data Science for Non-Life Insurance (D0N55a)
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Introduction

We construct a tariff plan for a Belgian motor third-party liability (MTPL) portfolio by decomposing the pure premium into frequency and severity to model the annual claim count with a Poisson family and the average claim cost with a Gamma family. The generalised additive model (GAM) captures the non-linear effect of driver age and the spatial effect of the postal code through penalised smoothers, which is translated into an interpretable generalised linear model (GLM) by discretising the smoothers into bins representing the risk level. Moreover, we use the gradient boosting machine (GBM), a tree based machine learning method that is tuned by cross-validation. The two tariffs resulting from each model are then evaluated using four tools: to verify that total predicted premiums match the observed losses, Lorenz curves and Gini coefficients to measure how well each model separates high-risk policies from low-risk policies, actual versus expected plots to assess calibration across the premium range, and a double lift chart to directly compare the two models against each other. Finally, a safety loading is derived from a Monte-Carlo simulation of the aggregate portfolio loss using the Tail-Value-at-Risk at the 99% level. The GBM achieves slightly stronger risk discrimination, while the GLM derived from the GAM remains easy to interpret and implement.

Data and Exploratory Analysis

The Portfolio

The data set contains 163,657 MTPL policies of a Belgian insurer. Each record describes one policy through a set of a priori covariates or risk factors, the realised claim experience and the exposure. The variables used in the analysis are listed in Table 1. The continuous driver age `AGEPH` and the spatial factor `CODPOSS` have to be accounted for in a different way, all remaining risk factors are categorical. The postal code is summarised as a pair of continuous coordinates, namely the longitude and latitude of the centroid of the corresponding municipality coming from the file `inspost.xls`, allowing location to be accounted for as a smooth spatial effect.

Table 1: Variables used in the analysis.

Variable	Description	Role / type
<code>nbrtotc</code>	Number of claims	Frequency response (count)
<code>chargetot</code>	Total claim amount (EUR)	Severity input
<code>duree</code>	Exposure-to-risk (years)	Offset / weight
<code>lnexpo</code>	$\log(\text{duree})$	Offset term
<code>AGEPH</code>	Age of the policyholder	Continuous rating factor
<code>CODPOSS</code>	Belgian postal code	Spatial key (<code>long</code> , <code>lat</code>)
<code>agecar</code>	Age of the vehicle	Categorical
<code>sexp</code>	Gender of the policyholder	Categorical
<code>fuelc</code>	Fuel type	Categorical
<code>split</code>	Premium payment split	Categorical
<code>usec</code>	Use of the vehicle	Categorical
<code>fleetc</code>	Fleet indicator	Categorical
<code>sportc</code>	Sports-car indicator	Categorical
<code>coverp</code>	Coverage level	Categorical
<code>powerc</code>	Engine power class	Categorical

The severity response is the average cost per claim, $s_i = \text{chargetot}_i / \text{nbrtotc}_i$, which is defined for policies that report at least one claim. For every categorical rating factor, the reference level is set to the level with the most exposure, so that the GLM intercept represents the baseline prediction for the most exposed class.

Exposure to risk and the offset

Not every policy is observed for a full year. The exposure d_i (**duree**) measures the fraction of the year during which policy i was actually at risk. Exposure can be incorporated into a claim count model in two ways: either the annualised count N_i/d_i is used as response with d_i as a likelihood weight, or the count N_i is kept as response and $\log d_i$ is added to the linear predictor as an offset:

$$N_i \sim \text{Poisson}\left(\exp(\log d_i + \mathbf{x}_i^\top \boldsymbol{\beta})\right) \quad (1)$$

The empirical histogram of **duree** (Figure 1) shows that the vast majority of policies are exposed for almost exactly one year. The offset $\log(d_i)$ scales the predicted count by exposure because a policy observed for half a year should be expected to produce half the claims of a full year policy.

Distributions of the responses

Figure 1 displays the distribution of the claim count, of the exposure and of the average claim cost. Almost 89% of contracts report zero claims, and nearly all of the remainder report a single claim; only about 1% report two or more, up to a maximum of five. The count distribution is strongly right-skewed with a large proportion of zero claims, which is naturally accommodated by the Poisson family when the mean claim frequency is small. The average claim cost is heavily right-skewed with a long tail, or a small number of very large claims, which motivates the Gamma family for severity.

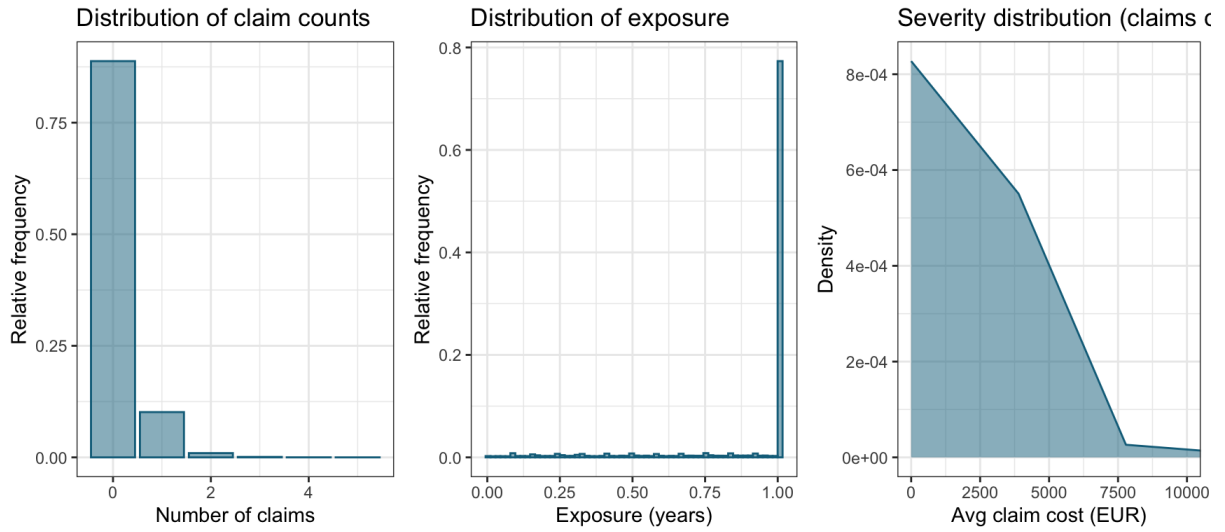


Figure 1: Distribution of the claim count, the exposure and the average claim cost (claims only, horizontal axis truncated at 10 000 EUR for readability).

Rating factors

Figure 2 shows the relative frequency of every rating factor. The portfolio is dominated by petrol cars, male drivers, private use, vehicles aged 6–10 years and the basic MTPL cover. Several levels: fleet vehicles, sports cars, professional use and the highest power class, represent under 5% of the portfolio, meaning their corresponding GLM coefficients are estimated from very few observations and are therefore less reliable.

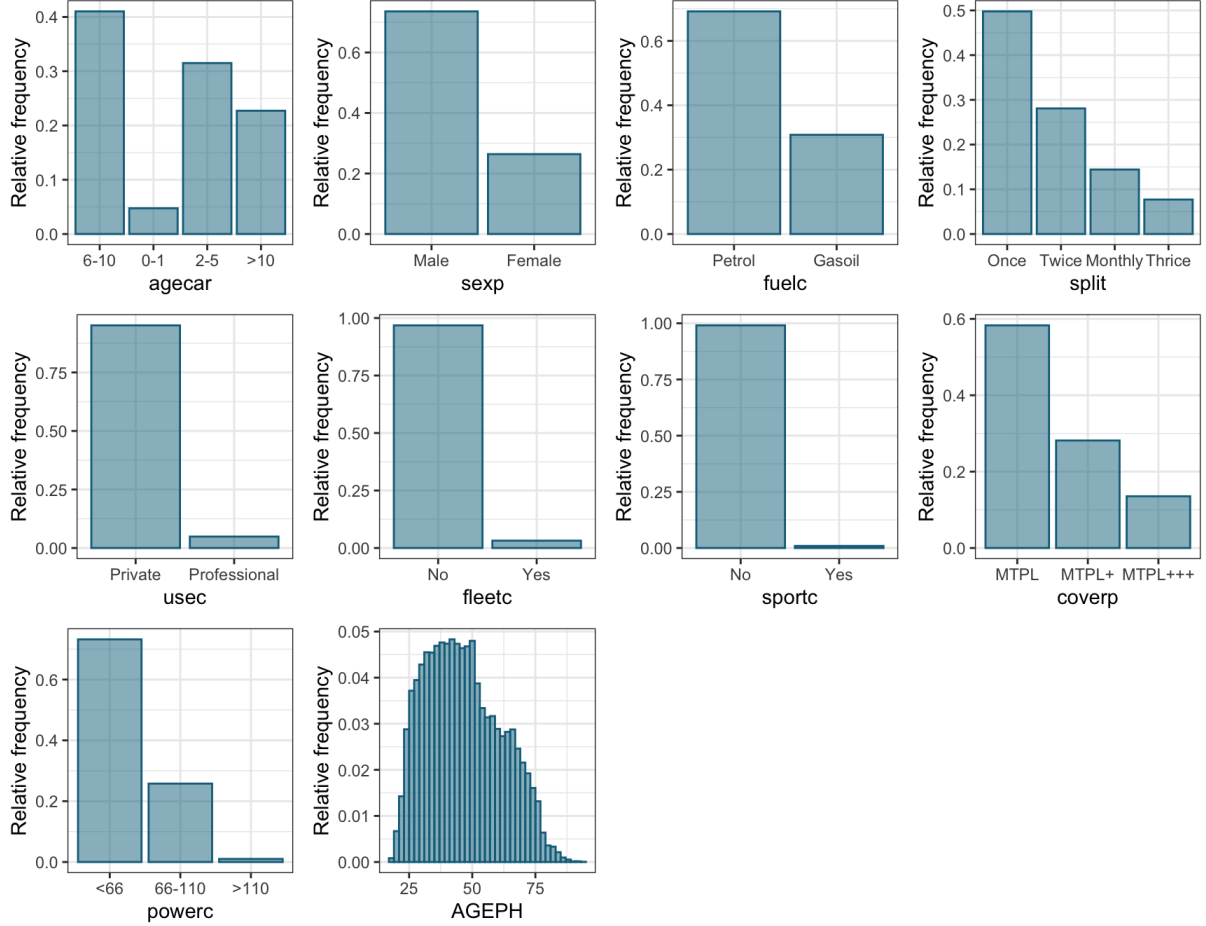


Figure 2: Relative-frequency distribution of the rating factors.

We compute the empirical claim frequency of each risk factor as the ratio of the total number of claims to the total exposure. Figure 3 confirms the expected ordering: coverage level, engine power and vehicle age separate the risk most clearly.

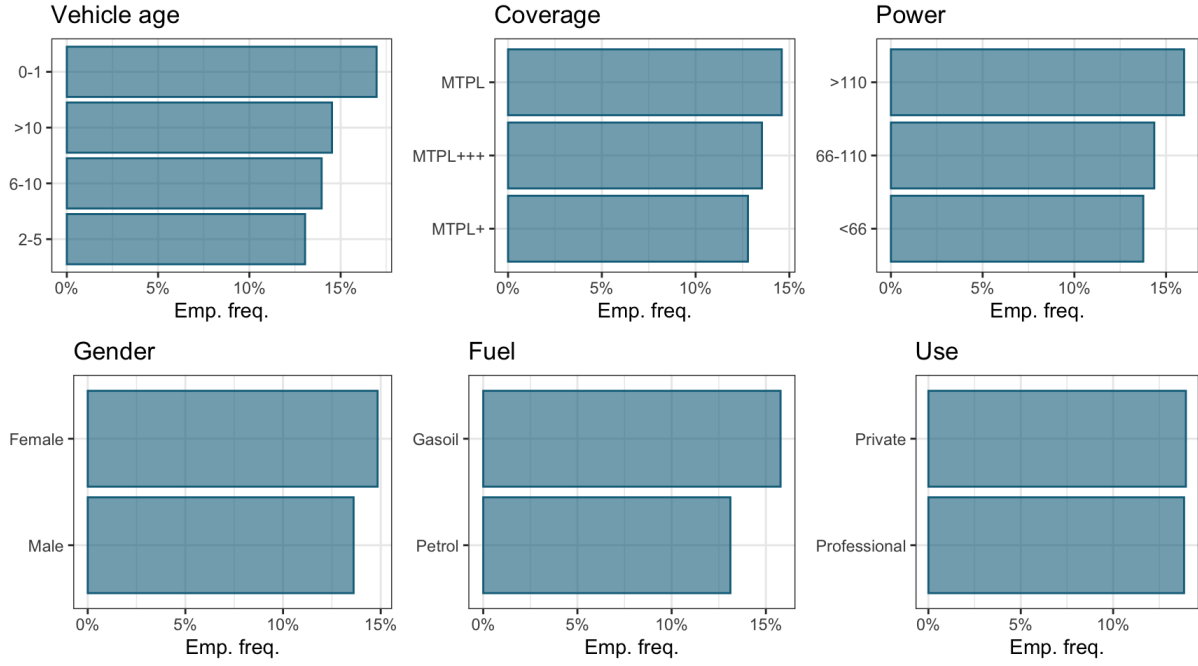


Figure 3: Empirical claim frequency by rating factor.

The age of the policyholder

Driver age is the most informative continuous rating factor. Figure 4 plots the empirical frequency against $AGEPH$: the relationship is non-linear. Claim frequency is high for young drivers, falls through the twenties and thirties, flattens out in middle age and slightly increases for the oldest drivers. A linear term in $AGEPH$ would be unable to reproduce this shape. It is therefore necessary to include the smooth term $s(AGEPH)$ introduced in the GAM in the next section. It is important to note that the 100%+ point and the 0% points are the result of having a single or very few policyholders at the extreme ages, so the empirical frequency is unreliable there and should not be relied on. We focus on the pattern.

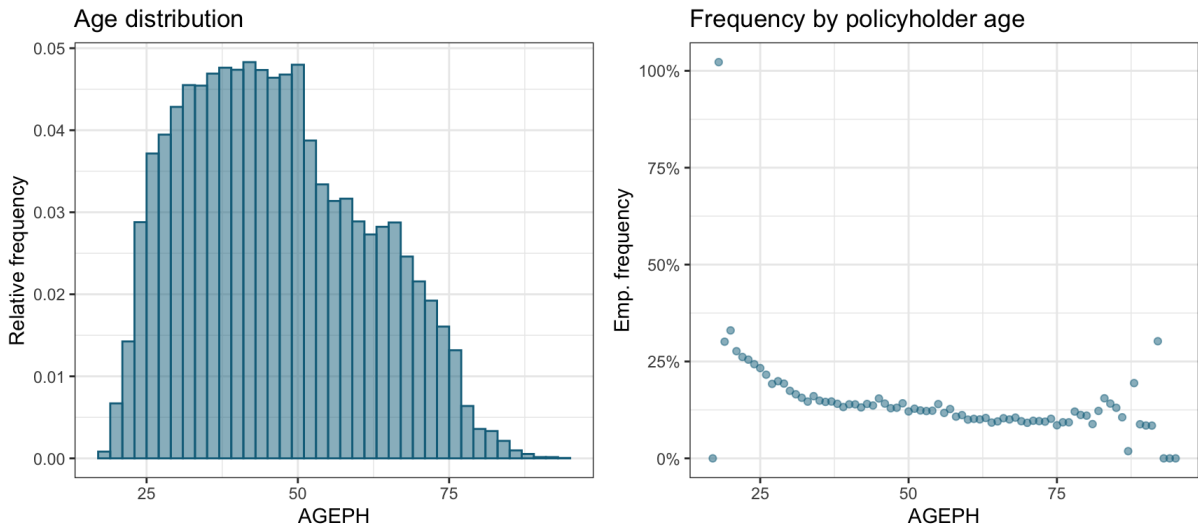


Figure 4: Age of the policyholder: histogram (left) and empirical claim frequency by age (right).

Train and test split

The data are split into a training set (80%) and a test set (20%) with a fixed random seed. All models are estimated on the training set and all diagnostics later on are computed on the test set. The severity models are estimated on the subset of the training set with at least one claim.

Frequency Modelling

The response is the claim count `nbrtotc`.

The Poisson GLM

Let N_i be the number of claims of policyholder i , with exposure d_i and rating factor vector $\mathbf{x}_i$ obtained by dummy encoding the categorical variables. The Poisson regression model assumes the counts to be independent with:

$$N_i \sim \text{Poisson}(d_i \lambda_i), \quad \lambda_i = \exp(\beta_0 + \sum_{j=1}^p \beta_j x_{ij}) = \exp(\mathbf{x}_i^\top \boldsymbol{\beta}) \quad (2)$$

The quantity $\mathbf{x}_i^\top \boldsymbol{\beta}$ is the linear predictor or risk score and the logarithm is the canonical link of the Poisson family. Because the link is the logarithm, the predicted frequency is a product of the baseline (e^{β_0}) multiplied by one factor per covariate. Each categorical covariate enters as dummy variables (0 or 1), so only the dummies that are equal to 1 contribute a multiplicative factor to the prediction. The dummies that are 0 drop out because $e^{\beta_j \cdot 0} = 1$.

$$\lambda_i = e^{\beta_0} \prod_{j: x_{ij}=1} e^{\beta_j}, \quad (3)$$

where e^{β_0} is the frequency of the reference class and e^{β_j} is the relativity of level j . A relativity above one increases the risk relative to the reference, a relativity below one decreases it. The coefficients are estimated by maximum likelihood. With k_i the observed count, the likelihood is:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^n \exp(-d_i \lambda_i) \frac{(d_i \lambda_i)^{k_i}}{k_i!}, \quad (4)$$

and $\hat{\boldsymbol{\beta}}$ maximises L . The relevance of each coefficient is determined by a Wald test of $H_0: \beta_j = 0$ using the statistic $Z = \hat{\beta}_j / \hat{\sigma}_{\hat{\beta}_j}$, which is approximately standard normal under H_0 by the asymptotic normality of the MLE. A large $|Z|$ means the estimate is many standard errors away from zero, so the covariate has a statistically meaningful effect. In the R output this corresponds to the `z value` column, with significance indicated by the stars. Each coefficient represents the log-frequency deviation from the reference level.

Model fit is measured by the deviance, the likelihood ratio test statistic of the fitted model against the saturated model (which fits every observation perfectly):

$$D(\mathbf{y}, \hat{\boldsymbol{\lambda}}) = -2 \ln \frac{L(\hat{\boldsymbol{\lambda}})}{L(\mathbf{k})} = 2 \sum_{i=1}^n \left[k_i \ln \frac{k_i}{d_i \hat{\lambda}_i} - (k_i - d_i \hat{\lambda}_i) \right] \quad (5)$$

The smaller the deviance, the closer the model is to the data. Each term in the sum compares the observed claim count k_i to the fitted count $d_i \hat{\lambda}_i$ in two ways: the first part, $k_i \ln(k_i / d_i \hat{\lambda}_i)$, penalises relative deviations in the log scale, while the second part, $(k_i - d_i \hat{\lambda}_i)$, penalises absolute deviations. Both terms are zero when the fitted count equals the observed count exactly, so the deviance equals zero for a perfect fit and grows as the model deviates from the data. The saturated model, which has one parameter per observation, achieves this perfect fit by construction and serves as the benchmark.

We further use information criteria, which balance goodness of fit against model complexity by penalising the number of parameters. A model with more parameters always fits better, but information criteria correct for this by adding a penalty, so that the selected model is the one that best generalises rather than the one that simply performs on the training data. The BIC penalises additional parameters more heavily than the AIC and is the criterion used to select the optimal number of spatial tariff classes.

From GLM to GAM

A plain GLM forces every continuous covariate to act linearly on the log-scale. However, insurance is rarely linear in driver age, and a GLM cannot represent smooth effects. We therefore use the generalised additive model to replace the linear terms of selected continuous covariates by unknown smooth effects:

$$\eta_i = g(\mu_i) = \beta_0 + \sum_{j=1}^p \beta_j x_{ij}^d + \sum_{j=1}^q f_j(x_{ij}^c) + \sum_{j=1}^r f_j(x_{ij}^s, y_{ij}^s), \quad (6)$$

where: x_{ij}^d are the discrete (categorical) covariates, x_{ij}^c are continuous covariates via smooth effects f_j , and (x_{ij}^s, y_{ij}^s) are spatial covariates.

The smooth functions f_j are represented as a sum of basis functions. Both the one-dimensional driver age smoother and the two-dimensional spatial smoother use thin-plate regression splines. To avoid overfitting, a wiggleness penalty is added to the log-likelihood. The smoothing parameters $\lambda_j \geq 0$ control the trade-off between fit and smoothness, where $\lambda \rightarrow 0$ yields a wiggly interpolation (low bias, high variance) and $\lambda \rightarrow \infty$ reduces the smooth to a straight line (high bias, low variance). The smoothing parameters are selected automatically by restricted maximum likelihood (REML).

Baseline frequency GAM

The baseline frequency model is a Poisson GAM with a log link that contains a thin plate spline of the spatial coordinates, a spline of driver age, the nine categorical rating factors and the exposure offset:

$$\text{nbrtotc} \sim s(\text{long}, \text{lat}) + s(\text{AGEPH}) + (9 \text{ categorical factors}) + \text{offset}(\text{lnexpo}) \quad (7)$$

The spatial smoother $s(\text{long}, \text{lat})$ captures geographic heterogeneity and $s(\text{AGEPH})$ captures the non-linear age effect identified in Figure 4.

Figure 5 maps the fitted spatial smoother evaluated per pair of postal-code centroid. Because the Poisson model uses a log link, the surface is expressed on the log-frequency scale. The positive values (dark) indicate locations where the expected claim frequency is above the portfolio average, negative values (light) indicate safer locations. The map shows a coherent spatial pattern where claim frequency is higher around the densely populated areas.

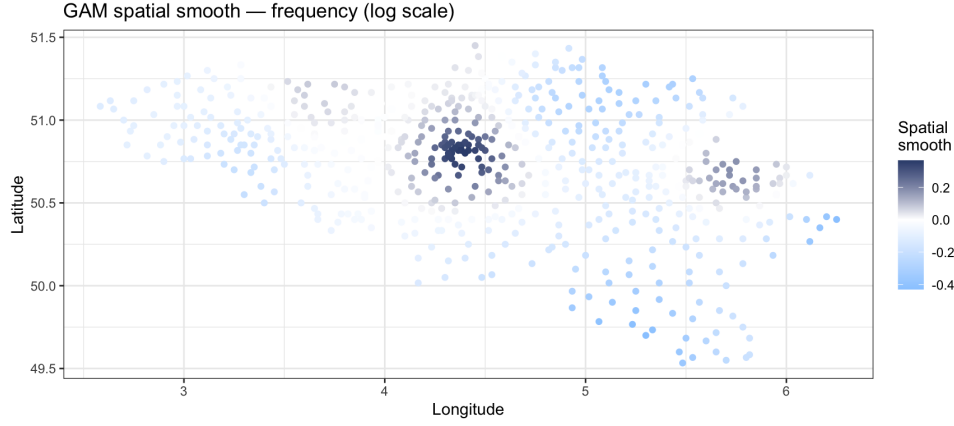


Figure 5: Fitted spatial smoother $s(\text{long}, \text{lat})$ of the baseline frequency GAM, on the log-frequency scale.

Figure 6 shows the estimated smooth $\hat{s}(\text{AGEPH})$ with its 95% confidence band. The curve also follows the empirical pattern: a steep decline in log-frequency from age eighteen to the early thirties, a flatter curve around the middle age, and a slight increase for the oldest drivers.

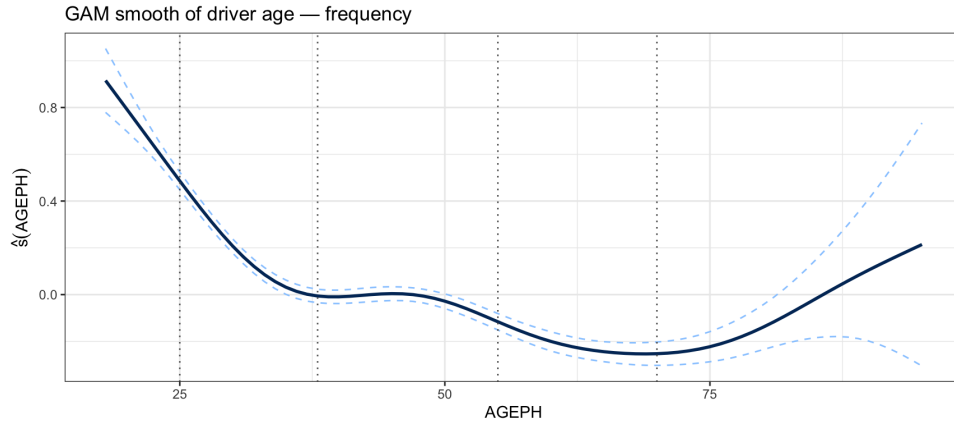


Figure 6: Fitted age smoother $\hat{s}(\text{AGEPH})$ of the baseline frequency GAM with 95% confidence band.

From smoothers to tariff classes

We discretise the two smoothers into classes representing a common level of risk, first the spatial smoother into geographical bins, then the age smoother into age bins, after which the model becomes a GLM.

Once the GAM is fitted, the spatial smooth is evaluated at every postal-code centroid. These predicted values are then grouped into k risk classes using Fisher’s natural-breaks algorithm, which places the class bins so that postal codes within the same class have similar predicted levels of risk. To choose the optimal number of classes, we refit the GAM for each candidate $k = 2, \dots, 7$, replacing the spatial smoother with the resulting discrete factor, and record the BIC. The k with the lowest BIC is selected. Figure 7 plots the BIC and AIC against k . The BIC is minimised at $k = 5$ classes.

Table 2: BIC and AIC by number of geographical bins

k	BIC	AIC
2	101,074.1	100,840.9
3	100,984.1	100,741.1
4	100,928.7	100,675.7
5	100,879.8	100,617.0
6	100,881.6	100,609.0
7	100,892.9	100,610.5

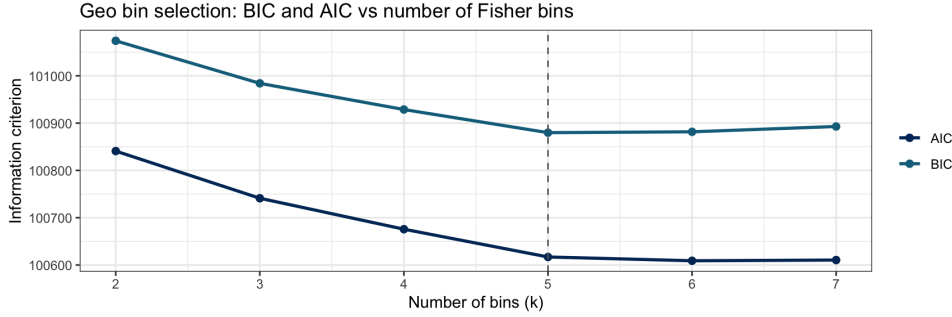


Figure 7: Selection of the number of spatial tariff classes: BIC and AIC against the number of Fisher classes. The dashed line marks the BIC-optimal value.

With the spatial classes fixed, the spatial smoother is replaced by the discrete factor `geo_freq` and the GAM is refitted:

$$\text{nbrtotc} \sim s(\text{AGEPH}) + (9 \text{ categorical factors}) + \text{geo_freq} + \text{offset}(\ln\text{expo}) \quad (8)$$

The age smoother used to define the age classes comes from the geo-adjusted model rather than from the original GAM. This is because the age smoother in the baseline model absorbs part of the geographic variation, since age and location are correlated. Once geography is held fixed as a constant factor, $s(\text{AGEPH})$ reflects the pure age effect.

We approximate the age smoother with a step function by fitting a regression tree to the fitted smooth values, weighted by the number of policyholders at each age. A regression tree partitions the age into cut points and predicts the average smooth value in each bin. The intervals are grown by recursive binary splitting, where each split aims to maximise the reduction in the loss function. The loss function is the mean squared error between each age's fitted smooth value and the average smooth value of its bin, measuring how well the step function approximates the original smooth curve. The size of the tree is governed by the complexity parameter `cp`, where the tree minimises the penalised criterion. The resulting age bins are:

$$\text{AGEPH} \in \{18.0, 31.5, 54.5, 95.0\},$$

i.e. three age classes $[18, 31.5)$, $[31.5, 54.5)$ and $[54.5, 95]$, with the middle class taken as the reference since it has the highest exposure.

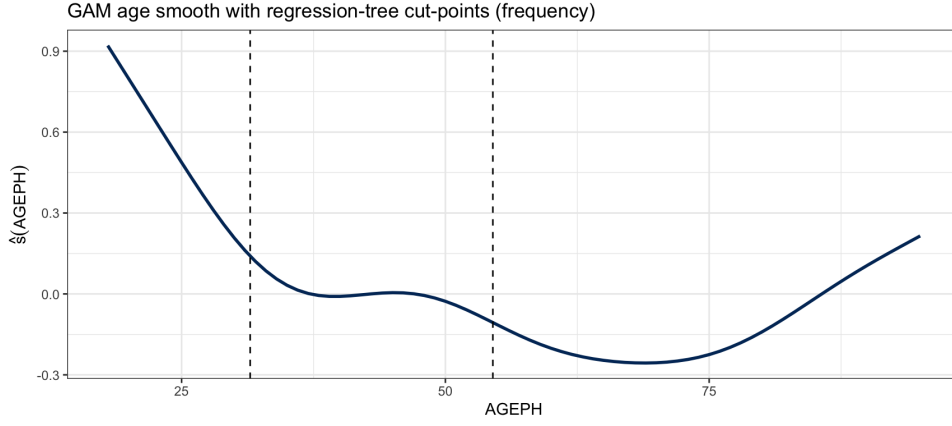


Figure 8: Age smoother of the geo-adjusted frequency GAM with the regression-tree cut-points (dashed) that define the age tariff classes.

Final frequency GLM

We replace the two smoothers by their corresponding binned factors `geo_freq` and `AGEPH_bin_freq` yielding the final GLM:

$$\text{nbrtotc} \sim (9 \text{ categorical factors}) + \text{geo_freq} + \text{AGEPH_bin_freq} + \text{offset}(\ln\text{expo}) \quad (9)$$

Every covariate is now categorical. Figure 9 shows the estimated coefficients with their 95% confidence intervals, coefficients whose interval excludes zero are statistically significant. The factors with the largest positive effect on frequency are the youngest driver-age class, the payment split (`split`), the highest engine-power class, the newest vehicles (`agecar` 0–1), and the highest `geo_freq` class. Coverage above the basic MTPL level and the oldest age class have the largest negative effects.

Table 3: Final frequency GLM

Coefficient	Estimate	Std. Error	z value	p -value	
(Intercept)	-2.1431	0.02297	-93.305	$< 2 \times 10^{-16}$	***
<i>Vehicle age (ref: 6-10)</i>					
agecar 0-1	0.2161	0.03721	5.808	6.34×10^{-9}	***
agecar 2-5	-0.0567	0.02019	-2.809	0.00497	**
agecar >10	0.0229	0.02085	1.100	0.27138	
<i>Gender (ref: Male)</i>					
sexc Female	0.0441	0.01803	2.446	0.01444	*
<i>Fuel (ref: Petrol)</i>					
fuelc Gasoil	0.2005	0.01720	11.663	$< 2 \times 10^{-16}$	***
<i>Payment frequency (ref: Once)</i>					
split Twice	0.1867	0.01885	9.905	$< 2 \times 10^{-16}$	***
split Monthly	0.3154	0.02432	12.969	$< 2 \times 10^{-16}$	***
split Thrice	0.3865	0.02726	14.179	$< 2 \times 10^{-16}$	***
<i>Use (ref: Private)</i>					
usec Professional	0.0424	0.03664	1.157	0.24708	
<i>Fleet (ref: No)</i>					
fleetc Yes	-0.0944	0.04822	-1.958	0.05018	.
<i>Sports car (ref: No)</i>					
sportc Yes	0.0080	0.08086	0.099	0.92095	
<i>Coverage (ref: MTPL)</i>					
coverp MTPL+	-0.1627	0.01938	-8.395	$< 2 \times 10^{-16}$	***
coverp MTPL+++	-0.1864	0.02755	-6.768	1.31×10^{-11}	***
<i>Engine power (ref: <66)</i>					
powerc 66-110	0.0845	0.01880	4.496	6.91×10^{-6}	***
powerc >110	0.3359	0.07676	4.377	1.21×10^{-5}	***
<i>Geo class (ref: [-0.066, 0.061))</i>					
geo_freq [-0.43, -0.21)	-0.2914	0.04319	-6.747	1.51×10^{-11}	***
geo_freq [-0.21, -0.066)	-0.1622	0.01962	-8.270	$< 2 \times 10^{-16}$	***
geo_freq [0.061, 0.22)	0.0216	0.02414	0.893	0.37179	
geo_freq [0.22, 0.37]	0.3351	0.02477	13.532	$< 2 \times 10^{-16}$	***
<i>Age class (ref: [31.5, 54.5))</i>					
AGEPH_bin [18, 31.5)	0.3874	0.01956	19.800	$< 2 \times 10^{-16}$	***
AGEPH_bin [54.5, 95]	-0.1950	0.02001	-9.745	$< 2 \times 10^{-16}$	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

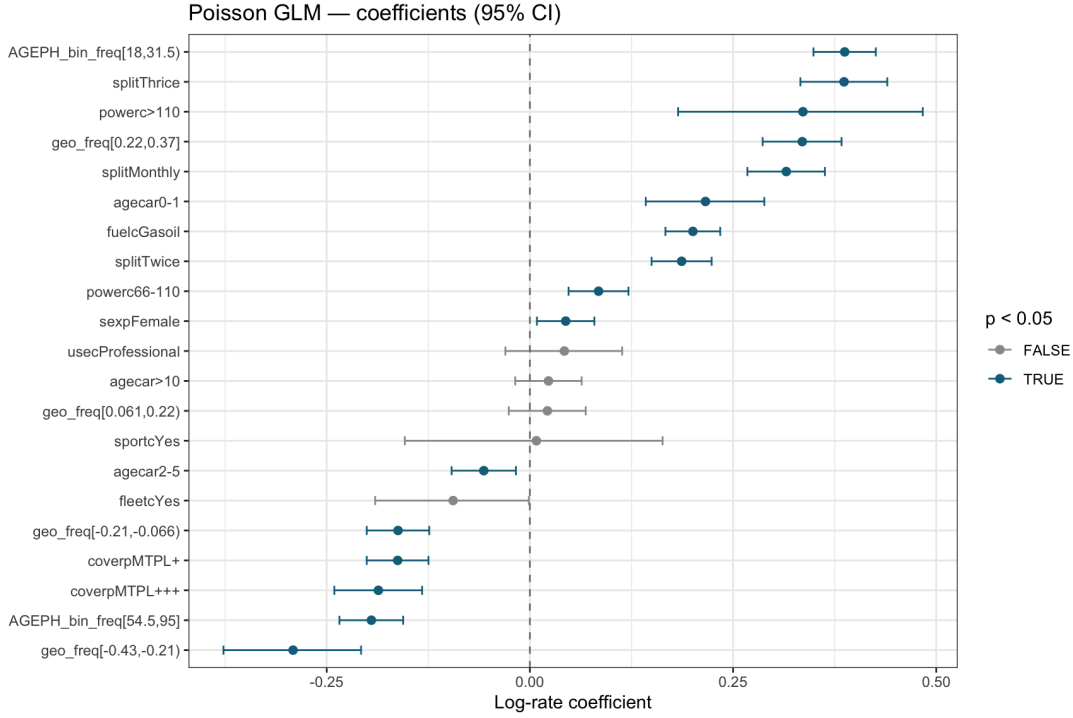


Figure 9: Estimated coefficients of the final Poisson GLM with 95% Wald confidence intervals. A coefficient is the logarithm of the multiplicative relativity in (3).

Gradient boosting for frequency

The GBM starts from a constant prediction, the single value that minimises the Poisson deviance over all training observations. It then builds the model over T iterations in a sequential way. At each iteration t , a random subsample of the training data is drawn without replacement (stochastic gradient boosting), which reduces variance and speeds up computation and for each observation in the subsample, the pseudo-residual is computed, which is the negative gradient of the Poisson deviance at the current prediction. This tells the algorithm in which direction and by how much the current fit is wrong for each observation. A regression tree is then fitted to these pseudo-residuals. The tree partitions the covariate space into regions, and within each region the optimal update is found by minimising the Poisson deviance. The update is multiplied by a shrinkage parameter before being added to the current fit, avoiding overfitting.

The GBM has several hyperparameters: the number of trees T (too many trees can cause overfitting), the interaction depth d (the number of splits per tree, controlling complexity), the shrinkage λ (the learning rate), the minimum node size, and the subsampling fraction δ . The number of trees, interaction depth, shrinkage and minimum node size are selected by five-fold cross-validation over a grid of 16 combinations, the subsampling fraction is fixed at $\delta = 0.7$.

Table 4: Optimal parameters for the frequency GBM selected by cross-validation

T	λ	d	Min. node size	δ	CV deviance	Best trees
3000	0.010	3	5000	0.7	0.7501	2948

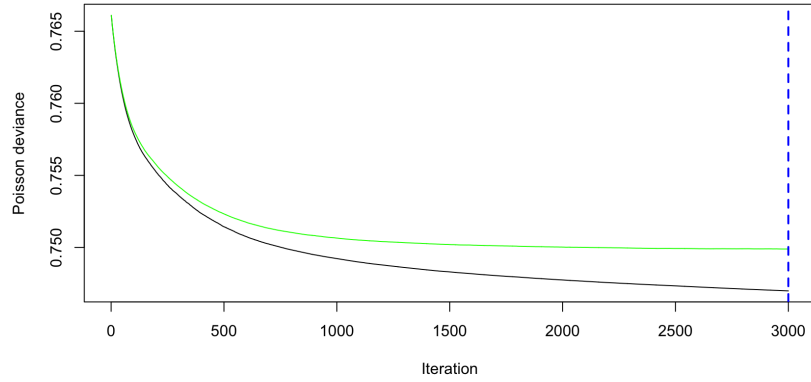


Figure 10: Cross-validation curve of the frequency GBM. The dashed line marks the number of boosting iterations that minimises the cross-validated Poisson deviance.

Variable importance is measured as the total reduction in the Poisson deviance achieved by splitting on that variable across all trees and all iterations. A variable that appears frequently and high in the trees contributes more to reducing the loss and therefore receives a higher importance score, as shown in Figure 11. Driver age dominates by a wide margin, followed by the payment split and the spatial coordinates (longitude and latitude), whereas fuel type, coverage and vehicle age contribute modestly. Engine power, use, and the sports-car and fleet indicators add little. This is aligned with the GLM, where driver age, the payment split and geography were also among the strongest effects.

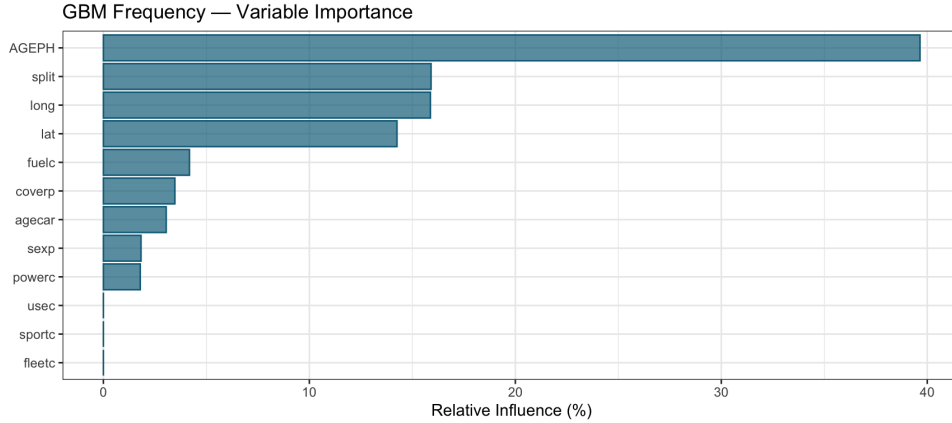


Figure 11: Relative variable importance of the frequency GBM.

PDP and ICE

The marginal effect of a single covariate is visualised with the PDP. As shown in Figure 11, driver age (`AGEPH`) is by far the most important variable, followed by the payment split and the spatial coordinates. We therefore focus the PDP and ICE analysis on `AGEPH` to examine how the GBM captures its non-linear effect. For ICE, we pick one specific policyholder at a time and keep all their characteristics fixed, but vary their age across the full range, resulting in the plot with the predicted frequency as age changes. For PDP, we take all those ICE curves and average them at each age value, which typically plots the effect of age on frequency across the whole portfolio. If all ICE curves are roughly parallel, it means that the age effect is similar for everyone. However, if they cross or diverge, the age effect depends on other characteristics or interactions. Figure 12 shows the ICE and PDP for driver age. The PDP follows the GAM age smoother, namely a steep decline through the twenties and thirties and a more gradual decline through middle age, but for the oldest drivers it flattens into a plateau rather than showing the slight increase seen in the GAM, since the trees do not place any additional splits here.

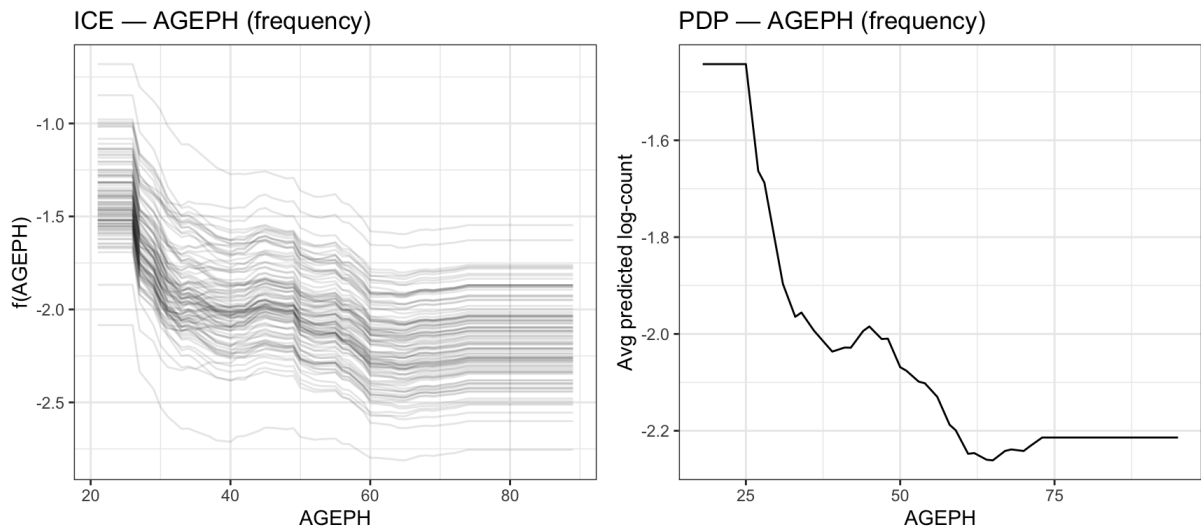


Figure 12: Individual conditional expectation curves (left) and partial dependence plot (right) of driver age in the frequency GBM.

Severity Modelling

The response now is the average claim cost $s_i = \text{chargetot}_i / \text{nbrtotc}_i$, estimated on policies with at least one reported claim.

The Gamma GLM and GAM

Claim amounts are strictly positive and right-skewed, so the severity component uses the Gamma family with a log link. Individual claim amounts Y_{ij} are assumed to be Gamma distributed with mean μ_i . The average cost $s_i = \frac{1}{n_i} \sum_j Y_{ij}$ is also Gamma distributed with the same mean μ_i , but with a shape parameter multiplied by n_i , meaning a policyholder with more claims has a more reliable average and should contribute more to the estimation. The Gamma GLM is fitted to s_i with n_i (**nbrtotc**) as weight, and the log link gives the same multiplicative tariff structure as for frequency.

We use the exact model setup as before. A baseline Gamma GAM,

$$\text{avg_claim} \sim s(\text{long}, \text{lat}) + s(\text{AGEPH}) + (9 \text{ categorical factors}), \quad \text{weights} = \text{nbrtotc} \quad (10)$$

The model is fitted by REML. It is important to note that no smoother is significant (spatial: $p = 0.301$, age: $p = 0.313$). However, the spatial smoother is still grouped into Fisher classes with k chosen by BIC over $k = 2, \dots, 7$, where the optimal is again $k = 5$. The severity GAM is estimated on the training claimants, far fewer than the training observations available for frequency, and the response is dominated by a small number of very large claims.

The spatial smoother is replaced by **geo_sev**, the GAM is refitted, and the age smoother of the geo-adjusted model is discretised with a regression tree, returning nine age classes: [18, 22.5), [22.5, 31.5), [31.5, 41.5), [41.5, 51.5), [51.5, 54.5), [54.5, 58.5), [58.5, 73.5), [73.5, 77.5), [77.5, 92], shown in Figure 13. Compared to frequency, we have nine bins now which reflects a stronger non-linearity in the age effect on claim costs.

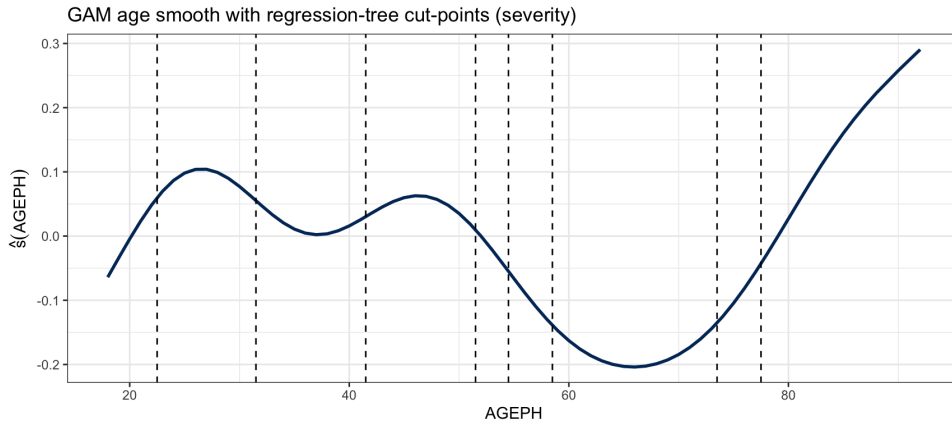


Figure 13: Age smoother of the geo-adjusted severity GAM with the regression-tree cut-points (dashed) that define the severity age classes.

The Gamma GLM is:

$$\text{avg_claim} \sim (9 \text{ categorical factors}) + \text{geo_sev} + \text{AGEPH_bin_sev}, \quad \text{weights} = \text{nbrtotc} \quad (11)$$

Table 5: Final severity GLM

Coefficient	Estimate	Std. Error	<i>t</i> value	<i>p</i> -value	
(Intercept)	7.4810	0.10930	68.442	$< 2 \times 10^{-16}$	***
<i>Vehicle age (ref: 6–10)</i>					
agecar 0–1	0.1259	0.15534	0.810	0.4178	
agecar 2–5	−0.0475	0.08304	−0.572	0.5677	
agecar >10	0.0494	0.08623	0.573	0.5667	
<i>Gender (ref: Male)</i>					
sexf Female	0.0316	0.07409	0.426	0.6701	
<i>Fuel (ref: Petrol)</i>					
fuelc Gasoil	−0.1601	0.07096	−2.256	0.0241	*
<i>Payment frequency (ref: Once)</i>					
split Twice	−0.1387	0.07809	−1.776	0.0758	.
split Monthly	−0.0192	0.09945	−0.193	0.8467	
split Thrice	−0.1109	0.11324	−0.980	0.3273	
<i>Use (ref: Private)</i>					
usec Professional	−0.0151	0.15154	−0.100	0.9204	
<i>Fleet (ref: No)</i>					
fleetc Yes	−0.3220	0.19930	−1.615	0.1062	
<i>Sports car (ref: No)</i>					
sportc Yes	−0.4680	0.33014	−1.418	0.1563	
<i>Coverage (ref: MTPL)</i>					
coverp MTPL+	−0.1495	0.08014	−1.865	0.0622	.
coverp MTPL+++	0.2240	0.11393	1.966	0.0494	*
<i>Engine power (ref: <66)</i>					
powerc 66–110	0.1369	0.07752	1.766	0.0773	.
powerc >110	0.5092	0.31331	1.625	0.1041	
<i>Geo class (ref: [−0.052, 0.1])</i>					
geo_sev [−0.43, −0.19)	−0.2296	0.11141	−2.061	0.0393	*
geo_sev [−0.19, −0.052)	−0.2454	0.08162	−3.006	0.0026	**
geo_sev [0.1, 0.26)	0.0964	0.10329	0.933	0.3508	
geo_sev [0.26, 0.45]	0.4150	0.11341	3.659	0.0003	***
<i>Age class (ref: [41.5, 51.5])</i>					
AGEPH_bin [18, 22.5)	−0.0363	0.21815	−0.166	0.8680	
AGEPH_bin [22.5, 31.5)	0.0113	0.09747	0.116	0.9074	
AGEPH_bin [31.5, 41.5)	−0.1129	0.09463	−1.193	0.2330	
AGEPH_bin [51.5, 54.5)	−0.1954	0.15876	−1.231	0.2184	
AGEPH_bin [54.5, 58.5)	−0.1101	0.15334	−0.718	0.4729	
AGEPH_bin [58.5, 73.5)	−0.2594	0.10918	−2.376	0.0175	*
AGEPH_bin [73.5, 77.5)	−0.2797	0.23851	−1.172	0.2410	
AGEPH_bin [77.5, 92]	0.0838	0.27985	0.299	0.7647	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

The severity GLM has a different result than frequency, where only three of the four geo classes, fuel type, coverage type MTPL+++ and one age class $[58.5, 73.5)$ are significant at the 5% level. This is consistent with the empirical data in Table 6, because while claim frequency decreases clearly with age, the mean claim cost also decreases with age (€1,812, €1,688, €1,317) but the median shows no consistent pattern (€567, €454, €627), reflecting the influence of a small number of large claims. Claim costs are therefore much harder to predict.

Table 6: Empirical claim frequency and average cost by age group

Age group	Emp. frequency	Policies with claims	Mean cost (EUR)	Median cost (EUR)
[18, 31.5)	0.2077	4,328	1,812.08	567.43
[31.5, 54.5)	0.1385	9,592	1,688.42	453.62
[54.5, 95]	0.1050	4,425	1,316.96	626.98

Gradient boosting for severity

The severity GBM follows the same setup as the frequency GBM. The natural loss for severity would be the Gamma deviance, but the Gamma distribution is not available in the CRAN version of the `gbm` package. We therefore model $\log(\text{avg_claim})$ with a Gaussian loss, the log-normal approach, which also yields strictly positive, right-skewed severity predictions. A Gaussian fit on the log scale estimates the mean of $\log(\text{avg_claim})$, so exponentiating the prediction directly returns the geometric mean rather than the expected severity. We therefore apply a log-normal smearing correction, multiplying the back-transformed prediction by $\exp(\hat{\sigma}^2/2)$, where $\hat{\sigma}^2$ is the variance of the log-scale residuals. The hyperparameters are tuned by the same 5-fold cross-validation, here over a grid of 36 combinations.

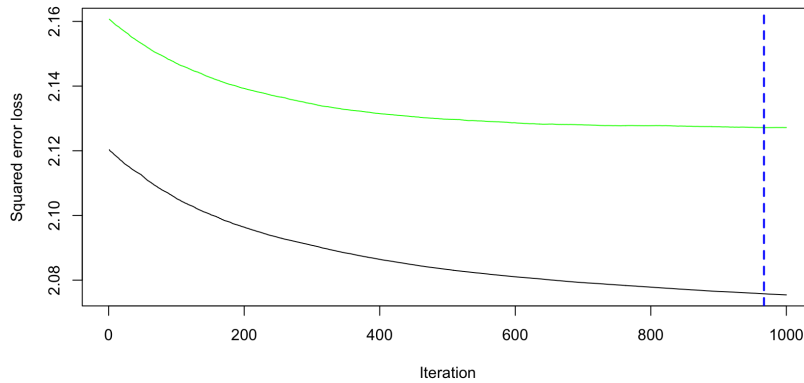


Figure 14: Cross-validation curve of the severity GBM.

Figure 15 shows the payment split as the most influential variable for severity, ahead of driver age, coverage and vehicle age, whereas for frequency the driver age dominated.

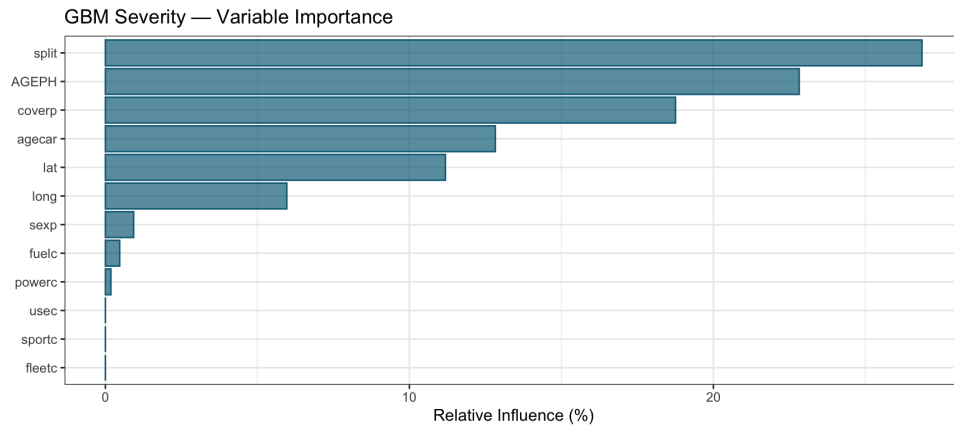


Figure 15: Relative variable importance of the severity GBM.

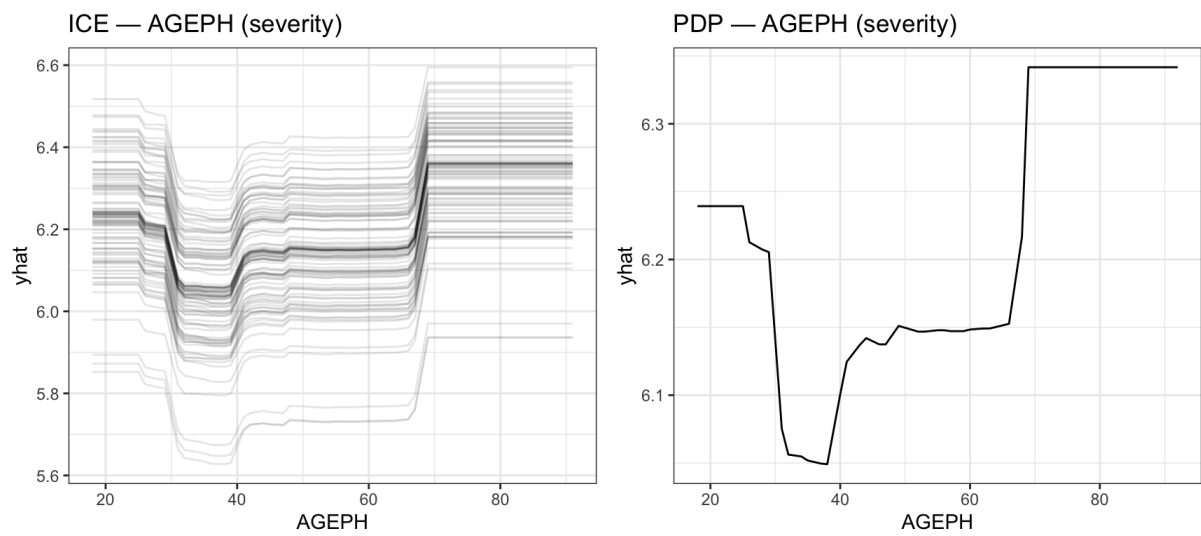


Figure 16: ICE curves (left) and partial dependence plot (right) of driver age in the severity GBM.

Model Evaluation and Comparison

In this section, we rely on the test set on which no estimation is done.

Portfolio balance check

We compare the total predicted cost against the total observed cost. The predicted-over-observed (P/O) ratio of the GLM tariff is given in the table below. A P/O ratio of 0.75 means the model predicts only 75% of the actual observed losses, which is a 25% underestimation. Since every policyholder is charged a premium regardless of whether they reported a claim, the predictions for severity are required for all policies in the test set, not only those with at least one claim. Moreover, for the values that fall outside the scope in the training set, the mean severity prediction is used. Hence, the P/O ratio of the GLM tariff is 0.75, indicating that the model underpredicts the total portfolio cost by 25%. The models are fitted on the training set and the P/O ratio is measured on the test set. This gap rather reflects the heavy tail of the claim cost distribution, where a few very large claims raise the observed total above the sum the severity model predicts.

Table 7: Portfolio balance check

Model	Predicted total (EUR)	Observed total (EUR)	P/O ratio
GLM (Poisson $\times$ Gamma)	6,139,735	8,191,228	0.75

Lorenz curve and Gini coefficient

The Lorenz curve measures the discriminating power, or the ability to charge more to the risks that actually cost more. We construct a Lorenz curve for each frequency model. Policies in the test set are ranked from lowest to highest predicted frequency. A model with no discriminative power would assign the same predicted frequency to every policy, meaning that the curve would lie on the dotted line. However, if the model separates low-risk policies from high-risk policies, then the curve will bow down.

The Gini coefficient is defined as twice the area between the diagonal and the Lorenz curve, which lies between zero (no discrimination) and one (perfect discrimination). The GBM has a slightly higher coefficient providing a stronger risk separation.

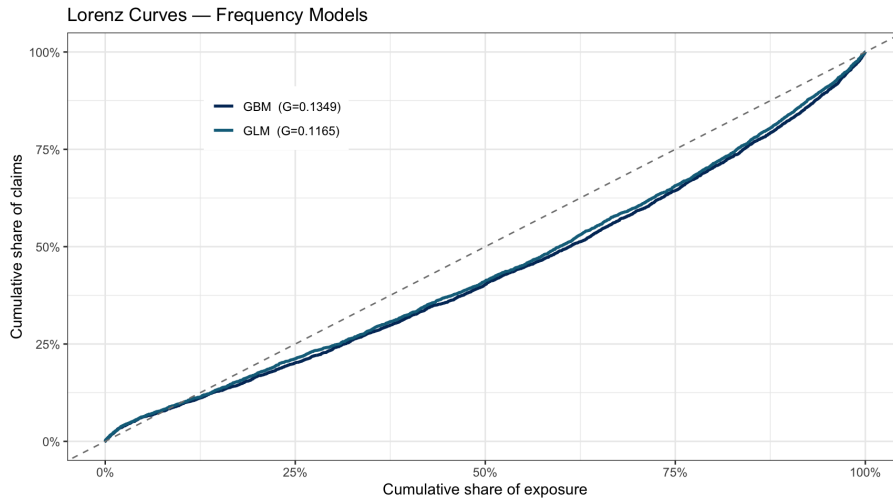


Figure 17: Lorenz curves of the frequency tariffs on the test set. The dashed diagonal is the line of no discrimination; a curve further from it indicates sharper risk separation.

Actual versus expected

The actual-versus-expected (AvE) plot checks whether the model's predicted premiums are too high or too low across the premium range. The policies in the test set are sorted by their predicted pure premium and grouped into 50 equal-sized bins, where for each bin the mean prediction is plotted against the mean observed cost. A perfectly calibrated model places every point on the 45° line.

The left panel of Figure 18 shows that most bins sit close to the diagonal, but there are a few outliers which is due to the heavy tail of the severity distribution. One or two very large claims can pull a bin's average observed cost far above its predicted level. The right panel takes this effect into account by winsorising the bin errors at their 99th percentile, capping the extreme tail. The outliers are reduced but many bins still sit above the diagonal, which is consistent with the portfolio balance check where the GLM pure premium underpredicts the observed costs across the premium range by 25%.

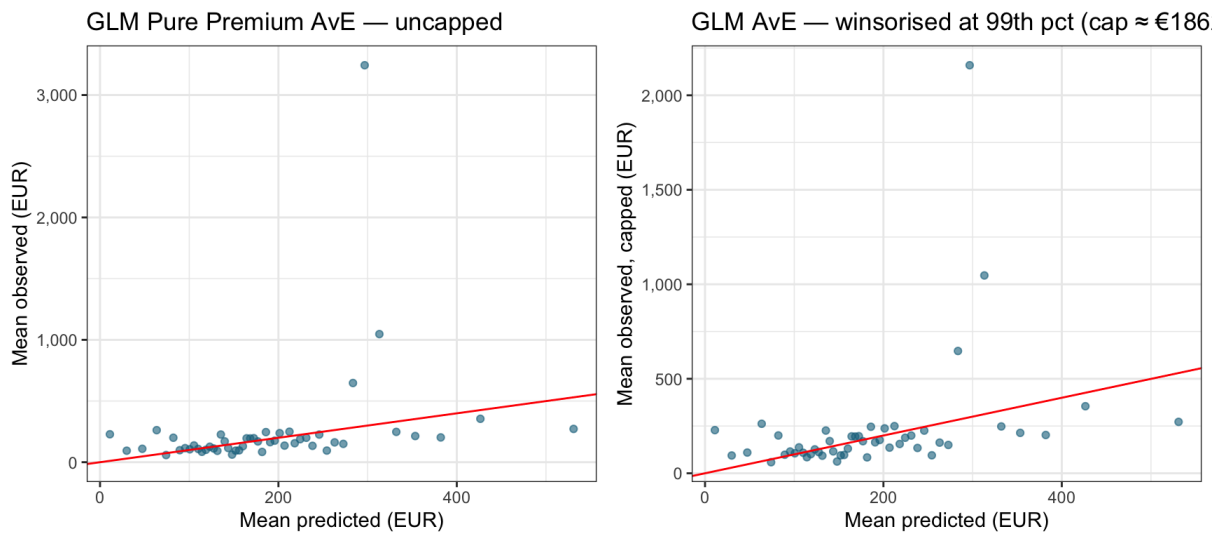


Figure 18: Actual-versus-expected plot of the GLM pure premium: raw (left) and winsorised at the 99th percentile of the absolute errors (right).

The same diagnostic can be done for severity to compare the two models directly, where we restrict the policies with at least one claim in the test set. The predicted severity is sorted and grouped into 20 equal-sized bins, where for each bin the mean predicted severity is plotted against the mean observed severity. Figure 19 shows the result for the GLM and the GBM tariffs.

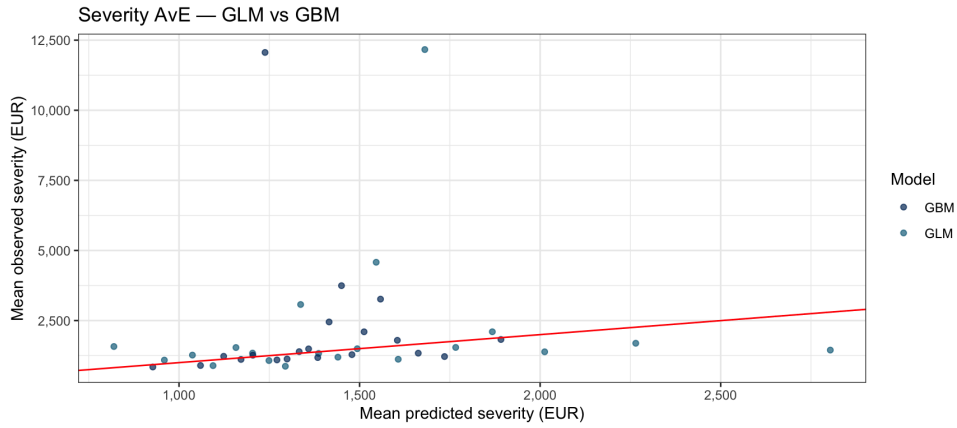


Figure 19: Severity actual-versus-expected plot: mean predicted against mean observed severity per bin, for the GLM and the GBM tariffs.

Double lift chart

The double lift chart compares the GLM and GBM against each other. Policies in the test set are sorted by the ratio of the GBM prediction to the GLM prediction and split into ten deciles. Decile one contains the policies where the GBM predicts a much lower frequency than the GLM, and decile ten contains those where it predicts a much higher frequency. We are interested in which model the observed claim rate closely follows.

In decile one, the GLM strongly overpredicts, while the GBM stays close to the observed claim rate. In decile ten, the GBM is almost perfectly aligned with the observed high-risk claim rate, whereas the GLM underpredicts it. In the middle deciles all three lines are close, so the models agree on medium-risk policies. Overall, the GBM tracks the observed claim rate more closely, especially in the extreme deciles.

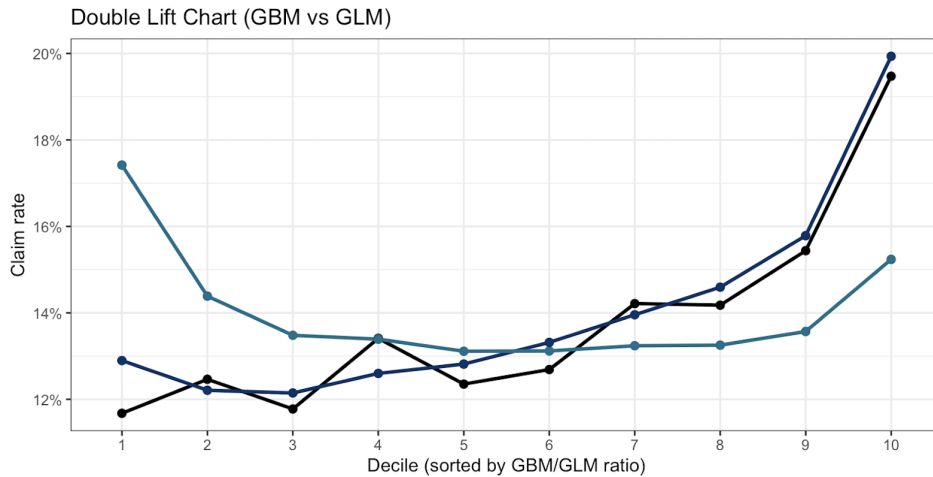


Figure 20: Double lift chart. Policies are sorted by the GBM/GLM ratio into ten deciles; each decile shows the observed claim rate against the average GLM and GBM predictions.

Policyholder profiles and the technical tariff

Three representative policyholder profiles are constructed to illustrate how the tariff responds to different risk combinations. We use the midpoint of each frequency age bin, approximately 25 for the youngest bin $[18, 31.5)$, 43 for the middle bin $[31.5, 54.5)$, and 75 for the oldest bin $[54.5, 95]$, such that each profile is guaranteed to fall in a distinct age class and the chosen age is representative of its group. The other factors are chosen to reflect risk, where the young driver (age 25) is assigned the high-risk level for every factor: a new vehicle ($\text{agecar} = 0-1$), high engine power ($\text{powerc} > 110$ kW), a sport car, professional use, and comprehensive coverage (MTPL+++). The experienced driver (age 43) takes on medium risk values: a mid-age vehicle (2–5 years), medium power (66–110 kW), no sport car, private use, and partial coverage (MTPL+). Finally, the senior driver (age 75) takes the safest level for every factor, an old vehicle (>10 years), low power (<66 kW), no sport car, private use, and basic coverage (MTPL).

Four factors are held constant across all three profiles at their reference level: sex (**Male**), fuel type (**Petrol**), payment split (**Once**), and fleet membership (**No**). Moreover, geography is fixed at the median spatial class for both frequency and severity.

The severity model contains only geography and policyholder age, but all other factors, such as vehicle age, engine power, sport-car indicator, use class, and coverage level appear only in the frequency model. Differences in pure premium across profiles therefore reflect primarily differences in predicted claim frequency, with a smaller age contribution on the severity side.

Table 8 reports the predicted frequency, severity, and pure premium for each profile under both models. We observe the highest GLM pure premium for the young driver which is driven by the combination of the highest frequency class, the largest expected claim cost, and the riskiest factor levels. The senior driver has a lower expected pure premium than the experience driver under the GLM. This reflects a decreasing age effect in the GLM where frequency and expected severity both decline as age increases beyond the youngest bin.

However, for the GBM the experienced driver is priced lower than the senior driver. This is driven by the GBM severity, likely reflecting the limited discriminative power of the log-normal for severity. Both models agree strongly on the young driver being the riskiest profile.

Table 8: Risk profiles for three representative policyholders

	Young Driver	Experienced Driver	Senior Driver
<i>Expected frequency (per year)</i>			
Freq — GLM	0.2619	0.1025	0.0987
Freq — GBM	0.2963	0.1286	0.1148
<i>Expected severity (EUR claim)</i>			
Sev — GLM	2,612.93	1,670.79	1,409.20
Sev — GBM	1,786.21	1,110.39	1,795.71
<i>Expected pure premium (EUR/year)</i>			
Pure prem — GLM	684.39	171.25	139.16
Pure prem — GBM	529.26	142.83	206.22

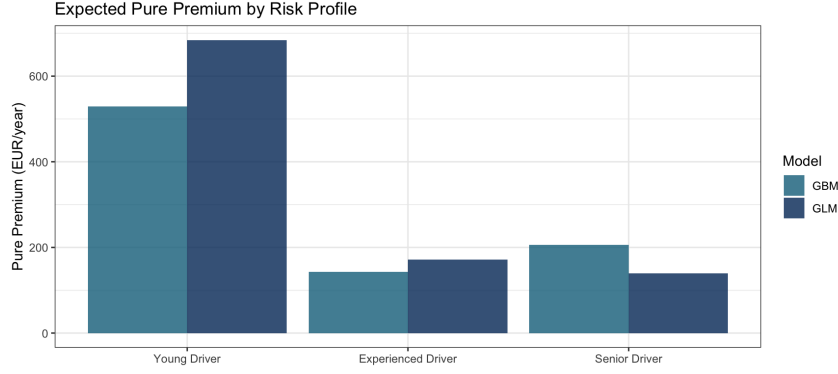


Figure 21: Expected pure premium of the three representative profiles under the GLM and the GBM tariff.

Safety Loading

The pure premium covers only the expected cost of claims. However, the realised aggregate loss can exceed the expected loss for which the insurer must add a safety loading to remain solvent with high probability. We use the TVaR to measure the portfolio tail risk at the 99% level (TVaR₉₉), which is the average aggregate loss in the worst 1% of years. The loading factor is

$$\theta = \frac{\text{TVaR}_{99}[S] - \mathbb{E}[S]}{\mathbb{E}[S]} \quad (12)$$

where S is the aggregate loss of the portfolio, and every policy's gross premium is its pure premium scaled by $(1 + \theta)$.

$$\theta = \frac{\text{TVaR}_{99} - \mathbb{E}[S]}{\mathbb{E}[S]} = \frac{7,332,976 - 6,137,314}{6,137,314} \approx 19.48\%.$$

Hence, every pure premium is scaled by a factor of 1.1948, so that the total loaded premium income across the portfolio equals the TVaR₉₉, providing a buffer sufficient to absorb extreme aggregate losses with 99% confidence.

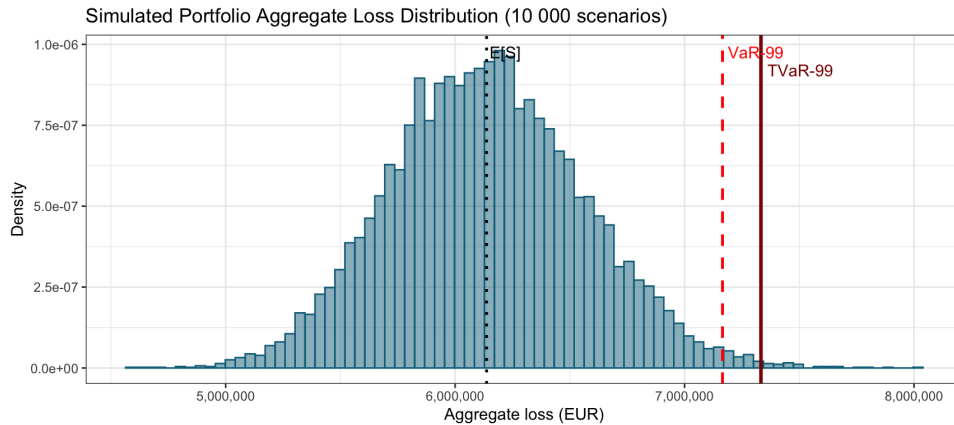


Figure 22: Simulated distribution of the portfolio aggregate loss (10 000 scenarios). The dotted line marks $\mathbb{E}[S]$, the dashed line VaR₉₉ and the solid line TVaR₉₉.

Table 9: Gross premiums with TVaR-99 safety loading ($\theta = 19.48\%$).

Profile	Pure Premium (EUR)	$\times (1 + \theta)$	Gross Premium (EUR)
Young Driver	684.39	$\times 1.1948$	817.72
Experienced Driver	171.25	$\times 1.1948$	204.61
Senior Driver	139.16	$\times 1.1948$	166.27